

The $SU(2)$ bridge midpoint in closed form: the curvature term is exact

Result, 2026-09-27. Proposition 6 of the series/parallel note described the midpoint of the Brownian bridge on $SU(2)$, the building block of the non-abelian parallel insertion, only to leading order in the heat time, with the curvature factor $1/h$, $h = (d/4) \cot(d/4)$, derived by Laplace's method and labelled as asymptotics. For the fundamental character the midpoint expectation is computable exactly. For a bridge of duration t from e to a group element at geodesic distance $\theta \in [0, 2\pi)$,

$$E \chi_{1/2}(m) = e^{-t/32} \left[2 \cos \frac{\theta}{4} - \frac{t}{2\theta} \sin \frac{\theta}{4} \right] \text{ up to image terms of relative size } \left(2 + \frac{32\pi^2}{t} \right) e^{-4\pi(2\pi-\theta)/t}$$

(Theorem 1). Expanded to first order it is exactly the prediction of Proposition 6(b), $2 \cos \frac{\theta}{4} [1 - \frac{t}{32} (1 + \frac{2}{h(\theta)})]$, because $\frac{t}{32} \cdot \frac{2}{h} = \frac{t}{4\theta} \tan \frac{\theta}{4}$. So the curvature term N2, the side-flux-dependent softening of the mid-face weight, is confirmed with no remainder of order t^2 in the $w = 0$ sector: the only correction to the displayed formula is the exponentially small winding of the bridge around the group. This is the first exact non-abelian statement about the parallel insertion in three dimensions, and it settles the fundamental-representation part of the uniform Laplace remainder that Proposition 7 of the series/parallel note assumes.

The ingredients are character orthogonality and Poisson summation; the computation is elementary and no novelty is claimed for it.

1. Setting

Group metric as in the series/parallel note §1: $C_2(j) = j(j+1)$, $SU(2)$ the three-sphere of radius 2, heat kernel $k_t = \sum_j (2j+1) e^{-tj(j+1)/2} \chi_j$ with $\chi_j(\theta) = \sin((2j+1)\theta/2) / \sin(\theta/2)$ at rotation angle θ (the geodesic distance from e). The midpoint m of the bridge of duration t from e to g has density $k_{t/2}(m) k_{t/2}(m^{-1}g) / k_t(g)$ with respect to Haar measure. Its geodesic midpoint $m_* = \exp(\frac{1}{2} \log g)$ has rotation angle $\theta/2$, so $\chi_{1/2}(m_*) = \sin(\theta/2) / \sin(\theta/4) = 2 \cos(\theta/4)$.

2. The exact formula

Theorem 1. Put $\Theta_{\pm}(\theta) = \sum_{w \in \mathbb{Z}} (\pm 1)^w e^{-(\theta - 4\pi w)^2/(2t)}$ and $\Xi_{\pm}(\theta) = \sum_{w \in \mathbb{Z}} (\pm 1)^w (\theta - 4\pi w) e^{-(\theta - 4\pi w)^2/(2t)}$. For $\theta \in (0, 2\pi)$,

$$E \chi_{1/2}(m) = e^{-t/32} \frac{2 \cos(\theta/4) \Xi_{-}(\theta) - \frac{t}{2} \sin(\theta/4) \Theta_{-}(\theta)}{\Xi_{+}(\theta)}. \quad (1)$$

Keeping only $w = 0$ gives the displayed formula of the summary.

Proof. Expand both heat kernels in characters. Character orthogonality in the form $\int \chi_a(m) \chi_b(m^{-1}g) dm = \delta_{ab} \chi_a(g) / d_a$, together with $\chi_{j_1} \chi_{1/2} = \chi_{j_1+1/2} + \chi_{j_1-1/2}$, gives

$$k_t(g) E \chi_{1/2}(m) = \sum_{j_1} \sum_{j_2=j_1 \pm 1/2} (2j_1+1) e^{-t[C_2(j_1)+C_2(j_2)]/4} \chi_{j_2}(g).$$

Write $n = 2j_1 + 1$. Then $C_2(j_1) + C_2(j_2) = \frac{1}{4}[n^2 + (n \pm 1)^2 - 2] = \frac{1}{2}(n \pm \frac{1}{2})^2 - \frac{3}{8}$, so the exponential is $e^{3t/32} e^{-t(n \pm 1/2)^2/8}$. Multiply by $\sin(\theta/2)$, reindex the $j_2 = j_1 - \frac{1}{2}$ sum by $n \mapsto n+1$, and write $s = n + \frac{1}{2}$ over $s \in \frac{1}{2} + \mathbb{N}$: the summand becomes $(s - \frac{1}{2}) \sin \frac{(s+\frac{1}{2})\theta}{2} + (s + \frac{1}{2}) \sin \frac{(s-\frac{1}{2})\theta}{2} = 2s \cos \frac{\theta}{4} \sin \frac{s\theta}{2} - \sin \frac{\theta}{4} \cos \frac{s\theta}{2}$, which is even in s . Hence

$$k_t(g) \sin \frac{\theta}{2} E \chi_{1/2}(m) = \frac{e^{3t/32}}{2} \left[2 \cos \frac{\theta}{4} A - \sin \frac{\theta}{4} B \right],$$

with $B = \sum_{s \in \frac{1}{2} + \mathbb{Z}} e^{-ts^2/8} \cos \frac{s\theta}{2}$ and $A = \sum_{s \in \frac{1}{2} + \mathbb{Z}} s e^{-ts^2/8} \sin \frac{s\theta}{2} = -2\partial_\theta B$. In the same way $k_t(g) \sin \frac{\theta}{2} = \frac{e^{t/8}}{2} A_0$ with $A_0 = \sum_{n \in \mathbb{Z}} n e^{-tn^2/8} \sin \frac{n\theta}{2}$. Poisson summation over the half-integers (which introduces the sign $(-1)^w$) and over the integers gives $B = \sqrt{8\pi/t} \Theta_-$, $A = (2/t) \sqrt{8\pi/t} \Xi_-$ and $A_0 = (2/t) \sqrt{8\pi/t} \Xi_+$. Divide. $\square$

The image terms. For $\theta \in (0, 2\pi)$ the pair $w = \pm 1$ in $\Xi_\pm$ has, relative to the $w = 0$ term $\theta e^{-\theta^2/(2t)}$, the size $e^{-8\pi^2/t} |2\theta \cosh(4\pi\theta/t) \mp 8\pi \sinh(4\pi\theta/t)|/\theta \leq (2 + 32\pi^2/t) e^{-4\pi(2\pi-\theta)/t}$, using $\sinh x \leq x \cosh x$. In Θ_- the same pair has relative size at most $2e^{-4\pi(2\pi-\theta)/t}$. The pairs $|w| \geq 2$ are smaller by a further factor $e^{-16\pi^2/t}$ up to polynomial factors. This is the large-field term N4, the winding of the bridge around the group, and it is negligible uniformly on $\theta \leq \theta_0 < 2\pi$.

3. Comparison with Proposition 6

Proposition 6(b) predicts that $m = m_* e^\xi$ with ξ Gaussian of mean zero and covariance $\frac{t}{4} \text{diag}(1, \frac{1}{h}, \frac{1}{h})$ in the frame (axis, transverse). In the spin- $\frac{1}{2}$ representation $(\xi \cdot T)^2 = -\frac{1}{4} |\xi|^2$, so $E D^{1/2}(e^\xi) = 1 - \frac{1}{8} E |\xi|^2 + O(t^2) = 1 - \frac{t}{32} (1 + \frac{2}{h}) + O(t^2)$, and $E \chi_{1/2}(m) = 2 \cos \frac{\theta}{4} [1 - \frac{t}{32} (1 + \frac{2}{h})] + O(t^2)$. The $w = 0$ part of (1) equals $2 \cos \frac{\theta}{4} e^{-t/32} [1 - \frac{t}{4\theta} \tan \frac{\theta}{4}]$, and $\frac{t}{32} \cdot \frac{2}{h(\theta)} = \frac{t}{16} \cdot \frac{\tan(\theta/4)}{\theta/4} = \frac{t}{4\theta} \tan \frac{\theta}{4}$. The two agree at first order, curvature factor included. As $\theta \rightarrow 0$ both give $2(1 - \frac{3t}{32})$, the flat value $E |\xi|^2 = \frac{3t}{4}$.

Formula (1) says more than the asymptotics. In the $w = 0$ sector the fundamental character of the midpoint is the product of an isotropic factor $e^{-t/32}$ and the exactly linear curvature correction $1 - \frac{t}{4\theta} \tan \frac{\theta}{4}$; there are no higher-order terms in t . The correction grows as $\theta \rightarrow 2\pi$, where $\tan(\theta/4)$ diverges and the midpoint becomes bimodal, and there the image terms take over. This is the non-abelian counterpart of the parity factor $\rho_t(\phi)$ of the $U(1)$ cube (Proposition 3 of the series/parallel note).

4. Consequence for STATE

The curvature term of the $SU(2)$ parallel insertion is now exact for the fundamental character. Two steps remain before Proposition 7 becomes a theorem for the isolated $SU(2)$ cube: the same computation for every spin J , since the cube's character sum $\sum_J d_J e^{-t_m C_2(J)/2} \text{tr}_J \prod_j E D^J(m_j^{\pm 1})$ involves all spins, and the matrix structure of $E D^J$ beyond its trace, which Proposition 6(a) reduces to one real diagonal matrix per cut face. The first follows the proof above with $\chi_{j_1} \chi_J = \sum_{|j_1-J|}^{j_1+J} \chi_{j_2}$ and is the next step on this line.